



Research Paper

High-strength and anti-biofouling nanofiber membranes for enhanced uranium recovery from seawater and wastewater

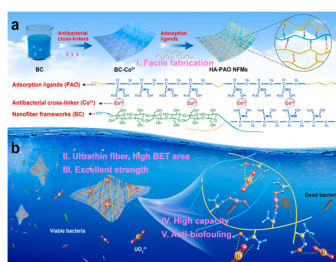
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HIGHLIGHTS

- A robust and anti-biofouling nanofiber membrane is designed by a facile method.
- The NFM exhibits ultrathin fiber structure, excellent strength, and high BET area.
- Significant inhibition zones show its good anti-biofouling ability.
- It can efficiently adsorb uranium from natural seawater and U-contained wastewater.
- It can be easily produced in a large scale and has potential in real applications.

GRAPHICAL ABSTRACT



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ABSTRACT

Ultrathin fibers can increase the contact area between adsorbents and seawater during the uranium extraction process; however, their construction usually aggravates the complex spinning technology and lowers their mechanical strength. Meanwhile, high strength and antifouling ability are essential for ocean adsorbents to withstand the complex natural environment and microbial systems. Herein, we design high-strength and anti-biofouling poly(amidoxime) nanofiber membranes (HA-PAO NFMs) via a supramolecular crosslinking. Bacterial cellulose supplies the NFMs with ultrathin fiber structure, and large amounts of adsorption ligands are immobilized on the framework via the crosslinking with antibacterial ions. Thus, different from other fibers, HA-PAO NFMs achieve ultrathin diameter (20–30 nm), high BET area ($51 \text{ m}^2 \text{ g}^{-1}$), and excellent mechanical strength (13.6 MPa). The uranium adsorption capacity reaches to 409 mg-U/g-Ads in the simulated seawater, 99.2% uranium can be removed from the U-contained wastewater, and the adsorption process can be observed by the naked eye due to the significant color changes. The inhibition zones indicate their excellent anti-biofouling ability, which contributes to 1.83 times more uranium extraction amount from natural seawater than the non-antifouling adsorbents. Furthermore, they display a long service life and can be large-scale prepared, and the HA-PAO NFMs have potential in the massive uranium recovery.

1. Introduction

The increasing use of fossil fuels is causing serious environmental

problems, and the exploitation of nuclear energy and other clean energy is imperative. Uranium is an important nuclear element and a strategic resource of the nuclear industry (Keener et al., 2020; Li et al., 2021; Yue

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et al., 2013). However, uranium resources on land are relatively scarce, only enough for about 100 years supplies (Abney et al., 2017). Fortunately, the ocean contains about 1000 times more uranium than on land, with up to 4.5 billion tons (Cheng et al., 2021; Wang et al., 2021b). However, the uranium extraction from ocean is challenging because of the low uranium concentration, complex natural environment, and microorganism system (Beltrami et al., 2014; Sun et al., 2021). To realize the large-scale exploitation, the adsorbents that simultaneously achieve good adsorption capacity, high mechanical strength, antifouling ability, and simple low-cost production are urgently required; however, they are difficult to obtain. Meanwhile, uranium is highly radioactive, and uranium recovery from nuclear wastewater is also important for environmental safety (Li et al., 2021). Therefore, the high-performance capture of uranium is vital not only for the energy strategy but also for the ecological protection.

However, the natural environment of ocean is complex, in order to resist the impact of waves and currents in the ocean, the adsorbents need a high-strength structure. Based on the chemical structure of the materials, the commonly used adsorbents can be broadly classified into gel adsorbents, membrane adsorbents, porous powder adsorbents, and fiber adsorbents. Among them, fiber adsorbents are the most studied and promising due to their chemical stability, corrosion resistance, and reusability (Shi et al., 2020; Zhang et al., 2022). Grafting of ligands onto the PE fibers, electro-spinning, and blow-spinning techniques are used to fabricate fiber adsorbents (Oyola and Dai, 2016; Wang et al., 2018, 2019; Zhang et al., 2019a). However, these methods usually produce micron-diameter fibers, and the specific surface area of the fibers is small, which is not favorable for the performance of uranium adsorption. To increase their specific surface area, two main strategies are usually adopted, that is, the construction of ultrathin or porous fibers (Wang et al., 2018; Xu et al., 2020). However, according to the results of previous studies, the above treatment processes always degrade the mechanical strength of the fibers, and the complex technologies make it difficult to realize large-scale production (Oyola et al., 2016; Shi et al., 2020). Fortunately, the natural world offers various opportunities for scientific research. Bacterial cellulose (BC) possesses a unique three-dimensional (3D) porous network structure and is a natural material consisting of ultrahigh aspect ratio fibers (Wu et al., 2016). Compared to plant-derived cellulose, BC has better purity, higher strength, more slender reticulation, and excellent hydrophilicity (Wu et al., 2016). Furthermore, it is affordable and readily available from industrial production. These unique properties inspired us to explore the applications of BC in uranium extraction. Therefore, we considered using BC to simultaneously achieve an ultrathin diameter, good specific surface area, and excellent tensile strength of the fiber adsorbents used for uranium extraction from seawater.

Notably, the concentration of uranium in ocean is very low, only 3.3 ppb, mainly in the form of $[\text{UO}_2(\text{CO}_3)_3]^{4-}$, and coexists with a variety of highly concentrated interfering ions (Das et al., 2016b; Li et al., 2020b; Yuan et al., 2019). Therefore, the adsorption ligand is an important factor affecting the performance of adsorbents. Among the many ligands, amidoxime (AO) is the most promising owing to its good affinity and selectivity for uranyl ions (Cheng et al., 2021; Haji et al., 2019). However, effective immobilization of AO on nanostructures is a challenge for AO-based adsorbents. Chemical modification, radiation-induced grafting, atom transfer radical polymerization, and ionic crosslinking have been used to bind ligands with matrix structures (Das et al., 2016b; Das et al., 2016c; Oyola and Dai, 2016; Oyola et al., 2016; Yan et al., 2020; Yue et al., 2013; Zhao et al., 2015). Among these methods, ionic crosslinking (Zn^{2+}) can be simply used to form 3D network hydrogels directly from the amidoxime anions (Yan et al., 2020). However, the Zn^{2+} -PAO gels obtained by the only ionic crosslinking are fragile (their mechanical strength is only 73.8 kPa) and they don't possess anti-biofouling properties. However, the antifouling ability is essential for the ocean adsorbents to endure the negative effects of marine microorganisms during uranium capture (Yuan et al., 2020b).

Fortunately, cobalt possesses a high antibacterial activity and is less toxic than silver. In addition, cobalt has been used to construct Co^{2+} -contained ZIF-67 and hybrid nanorods which displayed excellent anti-microbial activity (Aguado et al., 2014; Karthik and Thambidurai, 2017). Therefore, we considered using the Co^{2+} inspired ionic crosslinking and introducing BC frameworks to construct anti-biofouling and high-strength nanofiber membranes for the enhanced uranium recovery from seawater and wastewater.

Herein, we constructed high-strength and anti-biofouling poly(amidoxime) nanofiber membranes (HA-PAO NFMs) for uranium recovery via the bacterial cellulose assisted ionic crosslinking (Fig. 1). We added BC frameworks in the ionic crosslinking to increase the mechanical strength of the membranes. We selected poly(amidoxime) (PAO) as the adsorption ligands because of its high affinity for uranyl ions. We adopted Co^{2+} as cross-linkers owing to its good ability to inhibit bacterial growth and to effectively immobilize the adsorption ligands. As a result, the obtained HA-PAO NFMs exhibited an ultrathin fiber structure (20–30 nm) while high mechanical strength (13.6 MPa). A large number of PAO ligands were immobilized on the HA-PAO NFMs surfaces, giving them very excellent uranium uptake ability in the simulated seawater (409 mg-U/g-Ads) and uranium-contaminated wastewater (reduced uranium to 7.9 ppb). Significant inhibition zones demonstrated that bacteria can't grow on the HA-PAO NFMs surface, contributing to them 1.83 times more uranium adsorption amount than the non-antifouling samples. Furthermore, they exhibit a long service life and still can retain their structural stability after several adsorption-desorption cycles. The uranium adsorption process can be directly observed by the naked eye due to the significant color change of the NFMs. Meanwhile, the fabrication process is simple and economical compared with the traditional technologies for the fiber's construction. The high-performance properties, coupled with the simple and massive production, suggests the potential of HA-PAO NFMs in the real and large-scale uranium extraction from seawater and U-contaminated wastewater.

2. Results and discussion

2.1. Physicochemical characterization and structure analysis

The chemical component of HA-PAO NFMs was measured by FTIR spectroscopy. As shown in Fig. 2a, PAO had two peaks at 1656 cm^{-1} ($\text{C}=\text{N}$ stretching vibration) and 933 cm^{-1} ($\text{N}-\text{O}$ stretching vibration) (Liu et al., 2021); BC showed characteristic peaks at 1161 cm^{-1} ($\text{C}-\text{O}-\text{C}$ stretching) and 1031 cm^{-1} ($\text{C}-\text{O}$ stretching) (Yang et al., 2020; Huang et al., 2020b); the signals of both PAO and BC were present in the prepared HA-PAO NFMs, indicating the successful immobilization of PAO with BC nanofibers.

Furtherly, the PAO content in HA-PAO NFMs was calculated via the thermogravimetric analysis (TGA). As shown in Fig. 2b and Fig. S3, BC exhibited the weight loss at $250\text{ }^{\circ}\text{C}\sim 390\text{ }^{\circ}\text{C}$, attributing to the decomposition of cellulose skeletons. However, in the curve of HA-PAO NFMs, besides the mass losses before $338\text{ }^{\circ}\text{C}$ (ascribed to the decomposition of cellulose skeletons and amidoxime groups), an additional loss was present at $338\text{ }^{\circ}\text{C}\sim 560\text{ }^{\circ}\text{C}$, and this signal was attributed to the decomposition of main chain of PAO (to verify this result, the TGA of PAO also was performed, as shown in the blue line of Fig. 2b and Fig. S3). Therefore, the content of main chain of PAO in HA-PAO NFMs was 17.47%; meanwhile, based on the mass ratio of main chain to PAO (27:86), the PAO content in HA-PAO NFMs was calculated to be 55.65% (the detailed calculation process was shown in Supporting Information). This result was consistent with the data obtained by the weighting method.

Scanning electron microscopy (SEM) images showed the microstructure of the adsorbents at different magnifications. As shown in Fig. 2c and Fig. S2a-a', nanofibers with diameters of 20–30 nm were uniformly distributed in the BC, and these fibers were randomly twining forming a porous network, which could increase the specific surface area

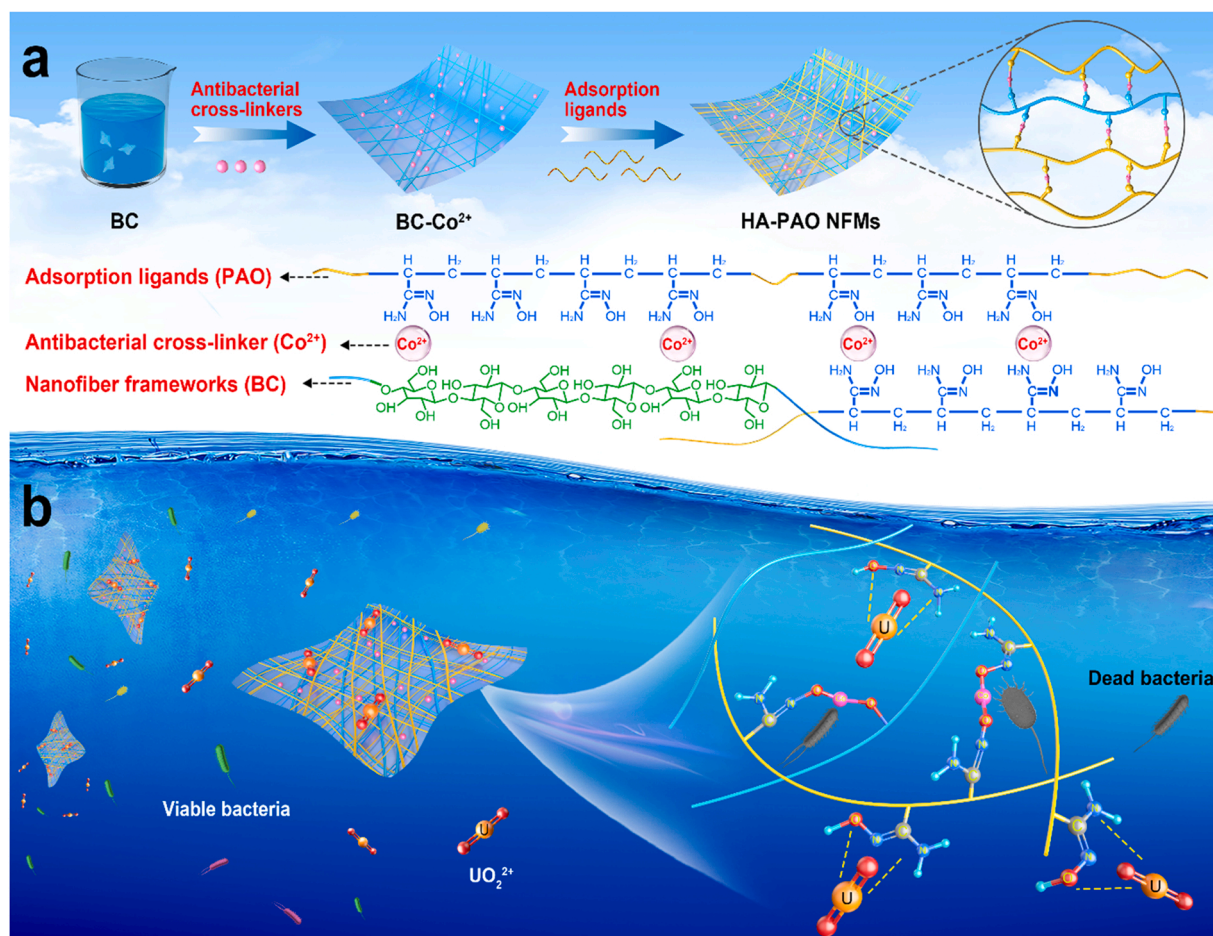


Fig. 1. a) Scheme of the construction of HA-PAO NFMs by the BC frameworks assisted ionic crosslinking. b) Illustration of their uranium adsorption mechanism and the antibacterial ability in the uranium recovery from seawater.

of NFMs. After PAO immobilization, many thin nanosheets were cross-linked on the nanofiber surfaces and formed a 3D porous structure with massive macropores, which could provide more active sites to bind uranyl ions. (Fig. 2e-f and Fig. S2b-b'). The immobilization of PAO resulted from chemical ionic crosslinking rather than physical accumulation. Energy dispersive spectrometry (EDS) mapping images of HA-PAO NFMs indicated the presence of Co²⁺ on the nanofiber membranes (Fig. S4a). The comparison of FTIR spectra of HA-PAO NFMs with PAO showed the strong interaction between PAO with nanofiber membranes: compared with the PAO, significant signal shifts for the amidoxime groups were observed in the HA-PAO NFMs (Fig. S4b).

The specific surface area and pore size information were investigated by the N₂ sorption-desorption experiments, and the data were analyzed by the Brunauer-Emmett-Teller (BET) and nonlocal density functional theory (NLDFT) methods. As the inset of Fig. 3a showed, the BET area of HA-PAO NFMs was calculated to be 51.02 m² g⁻¹, which was 1.70–26.85 times larger than those of the reported fiber-based adsorbents (Table S3) (Christou et al., 2019; He et al., 2020; He et al., 2021; Kaur et al., 2021; Li et al., 2020c; Oyola et al., 2016; Su et al., 2018; Wang et al., 2018), even was 4.02 times larger than that of the recently reported hierarchical porous ABP fibers (Xu et al., 2019). This result demonstrates that the ultrathin nanofibers really can increase the specific surface area of the adsorbents. Meanwhile, in addition to the macropores observed in the SEM images, the isotherms showed that numerous micropores with a pore size of 1.5 nm were present in the HA-PAO NFMs (Fig. 3a). The large BET area and the multistage pore structure contribute to the diffusion of seawater in the NFMs, accelerating the ion transport and increasing the uranium adsorption capacity of the

nanofiber membranes.

Hydrophilicity is a key parameter that affects the adsorption capacity. Therefore, the hydrophilicity of HA-PAO NFMs and the permeation rate of water droplets were assessed by dynamic water contact angle tests. Movie S1 showed that the water droplets quickly permeated into the nanofiber membranes, and the complete subsidence of water droplets into the membranes (the contact angle reached 0°) only needed 0.23 s (Fig. 3b). This rate was much faster than that of the only ion-crosslinked gels (Zn²⁺-PAO, 4 s) (Yan et al., 2020), and even much faster than the super-hydrophilic adsorbents (AOOBCH, 1.0 s) (Wang et al., 2021a). Besides the amidoxime ligands, the excellent hydrophilicity of the HA-PAO NFMs was also attributed to their nanofiber frameworks, again demonstrating the advantages of this nanofiber membrane compared to other fibers or membrane adsorbents. Furthermore, HA-PAO NFMs exhibited an excellent water capture capacity (51.67 ± 0.34 g g⁻¹). The excellent hydrophilicity and water uptake amount will enhance the diffusion of seawater into the adsorbents, and increase the uranium adsorption capacity.

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The construction of ultrathin nanofibers usually lowers the mechanical strength of adsorbents. However, the excellent strength is essential for the ocean adsorbents to withstand the impact of waves and currents in the real ocean environment. In order to investigate the mechanical strength of HA-PAO NFMs, the tensile experiments were performed in this work. The stress-strain curves demonstrated the good mechanical properties of the adsorbents (Fig. 3c). The tensile strengths of HA-PAO NFMs were 13.6 ± 0.9, 13.4 ± 0.6, 12.7 ± 0.8, and 12.4

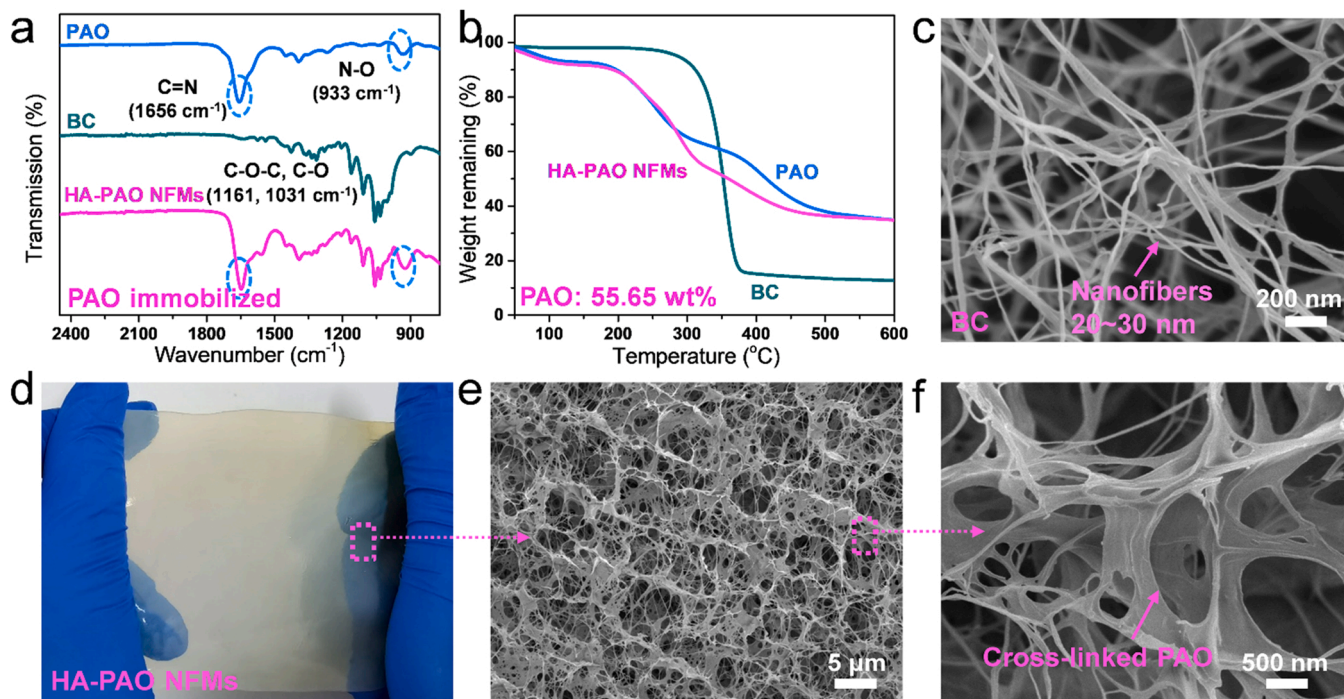


Fig. 2. a) FTIR spectra of the PAO, BC, and the prepared HA-PAO NFMs, indicating that the adsorption ligands (PAO) were immobilized on the nanofiber membranes. b) TGA curves of BC, PAO, and HA-PAO NFMs, analyzing the PAO content in the nanofiber membranes. c) SEM image of BC showing its nanofiber microstructures. d-f) Photograph and SEM images (with different amplifications) of the HA-PAO NFMs showing their macro-/micro-structures.

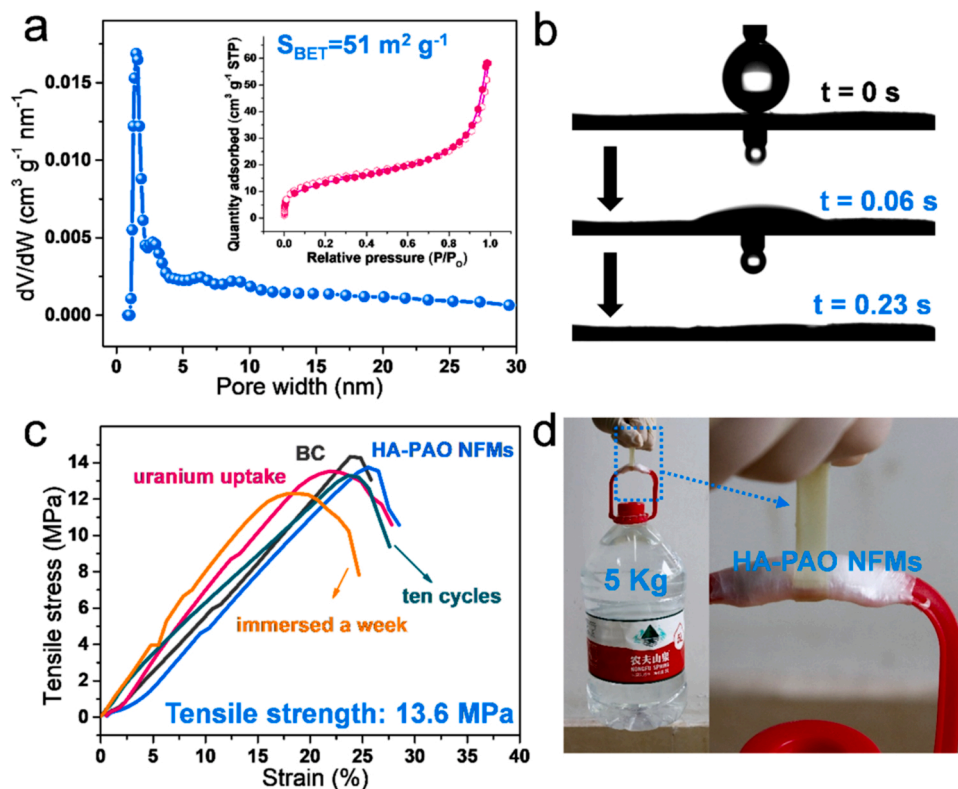


Fig. 3. a) BET area and pore size analyses of HA-PAO NFMs. b) The dynamic water contact angle test to determine the sink rate of water droplets in the HA-PAO NFMs. c) Stress-strain curves of the BC frameworks and the HA-PAO NFMs in their different states to investigate their mechanical properties in applications. d) The HA-PAO NFMs can withstand a 5 Kg weight, intuitively showing their excellent mechanical property.

± 1.0 MPa in the pristine state, post-adsorption state, after ten times of recycles, and after being immersed in the simulated ocean waves for a week, respectively, which quantitatively indicated their excellent mechanical properties. Meanwhile, the HA-PAO NFMs with a 0.18 mm thickness and a 10 mm width can withstand a 5 Kg weight without any breakage (Movie S2 and Fig. 3d), which intuitively demonstrates the excellent mechanical strength of the adsorbents. Different from other fibers, this HA-PAO NFM simultaneously achieves ultrathin diameter (20–30 nm), excellent specific surface area ($51.02 \text{ m}^2 \text{ g}^{-1}$), and high tensile strength (13.6 MPa), these advantages can enhance the uranium extraction efficiency and make the adsorbents work in the real ocean conditions to withstand the impact of ocean waves.

2.2. Uranium adsorption performances

The pH value can significantly affect the adsorption capacity of the adsorbents. Fig. S5 showed that the optimum pH of HA-PAO NFMs for uranium adsorption was pH 6 ($408 \pm 5.2 \text{ mg-U/g-Ads}$). Interestingly, the HA-PAO NFMs still exhibited good uranium capture capacities at acidic conditions (pH 3, $201 \pm 5.3 \text{ mg-U/g-Ads}$) and the pH range of natural seawater (pH 8, $293 \pm 5.8 \text{ mg-U/g-Ads}$). The above results demonstrate that the HA-PAO NFMs can be utilized in different uranium recovery environments including acidic nuclear waste and alkaline seawater.

The adsorption kinetics were investigated in the simulated seawater with different uranium concentrations. As displayed in Fig. 4a, the adsorption process was essentially completed at 36 h. The equilibrium adsorption capacities were $245 \pm 2.7 \text{ mg-U/g-Ads}$ and $409 \pm 6.8 \text{ mg-U/g-Ads}$ in 4 ppm and 8 ppm U-spiked simulated seawater, respectively.

The adsorption capacity increased with increasing uranium concentration, and the equilibrium adsorption amount increased to $526 \pm 2.0 \text{ mg-U/g-Ads}$ at the uranium concentration of 16 ppm. The uranium capture amount ($C_0 = 8 \text{ ppm}$, $409 \pm 6.8 \text{ mg-U/g-Ads}$) of the HA-PAO NFMs was much higher than those of the recently reported fiber-based adsorbents, membrane-based adsorbents, gel-based adsorbents, and even higher than those of the porous powder adsorbents (Table S4) (Bai et al., 2020; Chatterjee et al., 2015; Chen et al., 2020; Christou et al., 2019; Cui et al., 2020; Huang et al., 2020a; Kaur et al., 2021; Lin et al., 2022; Ma et al., 2019; Su et al., 2018; Sun et al., 2021; Wang et al., 2020; Xiong et al., 2019; Xu et al., 2019; Yuan et al., 2020a; Yuan et al., 2021; Zhong et al., 2020). The high uranium capture ability of HA-PAO NFMs was attributed to their efficient ligands, porous nanofiber structure, and excellent seawater penetration rate.

Furtherly, the distribution coefficient (K_d) was calculated to assess the affinity of the adsorbents toward uranyl ions, and the calculated K_d values were 81,307, 68,022, and $37,484 \text{ mL g}^{-1}$ for HA-PAO NFMs in the simulated seawater with uranium concentrations of 4, 8, and 16 ppm, respectively. The high K_d values also demonstrated the strong combination of HA-PAO NFMs with uranyl ions (Ma et al., 2015).

The adsorption behavior was in accordance with the pseudo-second-order models. As shown in Fig. 4b and Fig. S7, the correlation coefficients (R^2) were 0.9998, 0.9995, and 0.9982 for HA-PAO NFMs in the simulated seawater with different uranium concentrations, respectively, which were higher than those calculated using pseudo-first-order models (0.9356, 0.9742 and 0.9213, respectively). In addition, the calculated equilibrium adsorption capacities (251, 422, and 546 mg-U/g-Ads for the three uranium concentrations, respectively) according to

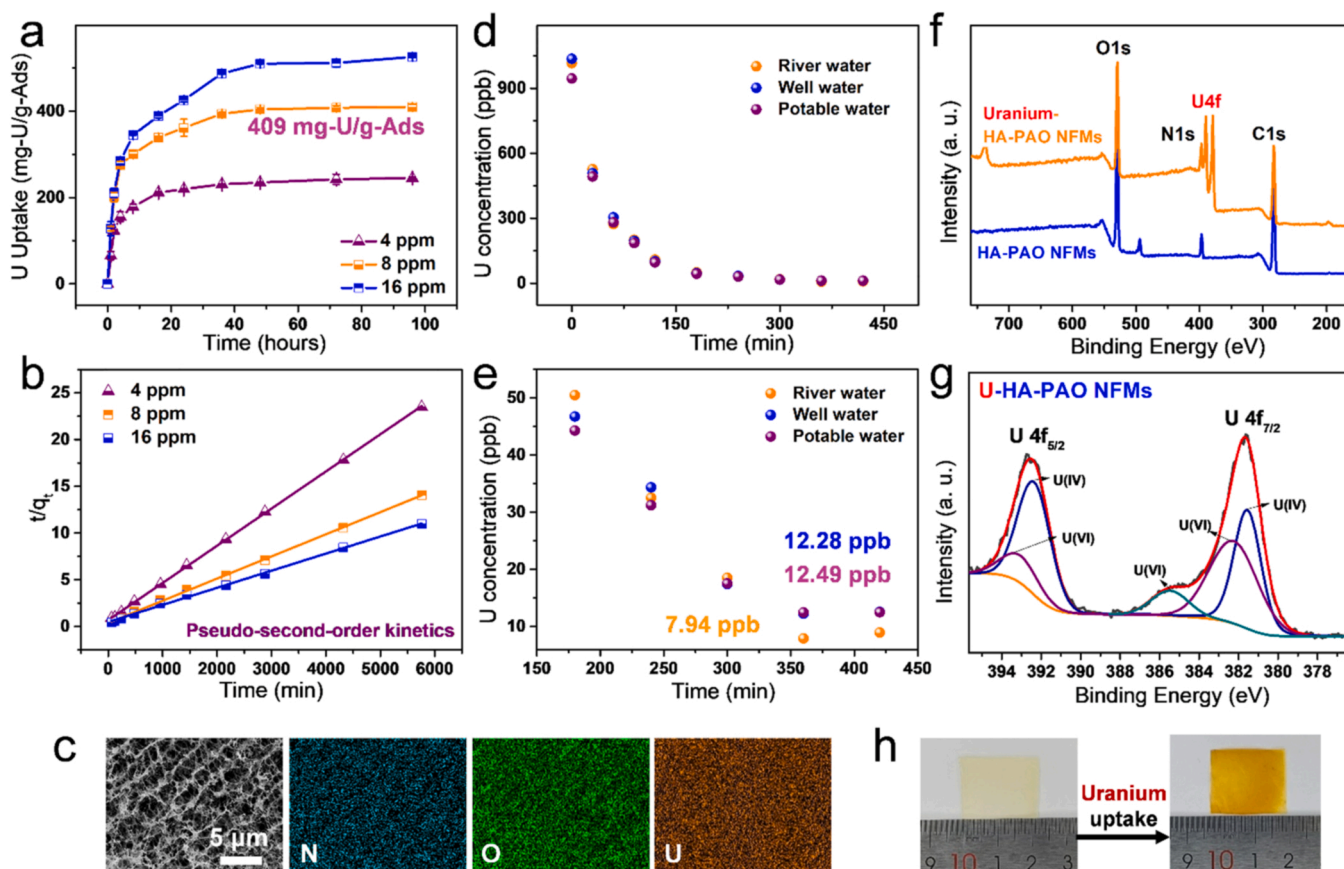


Fig. 4. a, b) Adsorption kinetics and the pseudo-second-order models of HA-PAO NFMs in the U-spiked simulated seawater. $V = 1000 \text{ mL}$, $m_{\text{ads}} = 5 \text{ mg}$. c) EDS mapping images of the HA-PAO NFMs after uranium capture. d, e) Removal kinetics of uranium by HA-PAO NFMs from various water samples. $V = 300 \text{ mL}$, $m_{\text{ads}} = 15 \text{ mg}$, $C_0 = 1000 \text{ ppb}$. f) XPS spectra of HA-PAO NFMs before and after uranium adsorption. g) High-resolution XPS spectra (U 4f) of the samples after uranium adsorption. h) Photos of the HA-PAO NFM before and after uranium uptake showing its significant color changes due to the uranium adsorption.

the pseudo-second-order models were closer to the experimental data (Table S1). The above results indicate that the adsorption process of HA-PAO NFMs for uranium is chemisorption rather than physical deposition.

Nuclear leakage from power plants and uranium mining may produce nuclear waste, which seriously harms the natural environment and life safety. Therefore, we investigated the uranium removal ability of the HA-PAO NFMs in the uranium-containing wastewater. Herein, three real water samples: river water, well water and potable water were used in the tests. Then, these water samples were intentionally contaminated with uranium (1000 ppb), the adsorbents (15 mg) were added to each water sample (300 mL) to adsorb uranium, respectively, and the residual uranium concentrations in the wastewater were measured by inductively coupled plasma mass spectrometry (ICP-MS). As shown in Fig. 4d and e, only need 6 h, the HA-PAO NFMs can remove 99.2%, 98.8%, and 98.7% uranium in these uranium-contained samples, respectively, and the uranium can be reduced to 7.94, 12.28 and 12.49 ppb for the wastewater samples, which are well below the discharge standards (30 ppb) of the U.S. Environmental Protection Agency for drinking water. These results showed the potential of HA-PAO NFMs in the removal of uranium.

The applications of adsorbents in the wastewater usually involve many interfering ions. Therefore, the competitive ion interference experiments were performed to further verify the potential of the HA-PAO NFMs. Interfering elements, including in lanthanides (Ce, Eu, La), radioactive ions (Cs, Sr), transition metal ions (Ni, Pb, Zn, Fe), and common ions (Na, Mg, K, Ca), were selected based on the references because they are likely to exist in the radioactive wastewater (Ahmad et al., 2022; Li et al., 2020a). Herein, the initial uranium concentration was 10 mg L^{-1} , and the concentrations of these interferences were 50, 20, 20, and 100 mg L^{-1} , respectively, which were 5, 2, 2, and 10 times of that of the U(VI). Fig. S8 showed that the HA-PAO NFMs still exhibited good uranium adsorption capacities in these interferences, their uranium capture amounts in the presence of interfering ions were 80.89–99.19% of that without the interferences, showing a high selectivity of HA-PAO NFMs towards U(VI). The prepared adsorbents have good potential for uranium recovery in the complex wastewater.

Furtherly, the selectivity experiments were performed by adding interfering ions to the natural seawater, and the adsorption capacities of HA-PAO NFMs for the different metal ions were measured by ICP-MS. As shown in Fig. S9, HA-PAO NFMs still exhibited good selectivity for uranium in the presence of many competing metal elements: the absorption capacity ($32 \pm 1.8 \text{ mg g}^{-1}$) for uranium was 2.28–108 times higher than those of other metal ions (except vanadium). The adsorption of amidoxime adsorbents for vanadium has been a difficult problem in uranium extraction for many years because of their inherent properties (Ivanov et al., 2017). These results indicate that the HA-PAO NFMs can uptake uranium from a large number of interfering ions and have good potential for uranium capture from complex seawater.

The mechanism of uranium adsorption by HA-PAO NFMs was investigated via EDS and X-ray photoelectron spectroscopy (XPS). As displayed in Fig. 4f, the XPS spectra of the post-adsorption samples exhibited U4f signals, which was not observed for the original HA-PAO NFMs, proving the successful chelation of uranyl ions with adsorbents. High-resolution spectra in U4f region further confirmed the presence of uranium in the post-adsorption nanofiber membranes. In addition, as shown in Fig. 4g, the U 4f spectrum could be further divided into five peaks, which were U(VI) $4f_{5/2}$ at 393.3 eV, U(IV) $4f_{5/2}$ at 392.5 eV, U(VI) $4f_{7/2}$ at 385.5 eV, U(VI) $4f_{7/2}$ at 382.4 eV, and U(IV) $4f_{7/2}$ at 381.6 eV, respectively (Feng et al., 2022; Mei et al., 2022; Zhang et al., 2019b). The EDS mapping images of the samples after uranium uptake showed the uniform distribution of U element, again proving the effectiveness of HA-PAO NFMs in capturing uranyl ions (Fig. 4c). Besides, significant color changes were observed for the nanofiber membranes after uranium adsorption, which also demonstrated the effective uptake of uranium by the HA-PAO NFMs (Fig. 4h). Meanwhile, Fig. S10

showed that the BC almost can not adsorb uranyl ions, and the PAO played the main role in the uranium capture for HA-PAO NFMs. Furtherly, the high-resolution spectra in O 1s region were detected to confirm the effective groups in the uranium adsorption process. As shown in Fig. S11, the O 1s spectra of HA-PAO NFMs before uranium adsorption were fitted into two peaks at 532.55 eV and 530.88 eV, which were attributed to the $\text{C}(\text{NH}_2)=\text{N}-\text{OH}$ and COOH groups, respectively. The COOH groups may come from the transition of part of amidoxime groups during the alkaline treatment in the preparation of the adsorbents (Das et al., 2016a, 2016d; Wang et al., 2018). After uranium adsorption, a new peak appeared at 531.55 eV (attributed to $\text{O}=\text{U}=\text{O}$), indicating the adsorption of uranium on the surface of HA-PAO NFMs. Meanwhile, the $\text{C}(\text{NH}_2)=\text{N}-\text{OH}$ signal of post-adsorption samples shifted to 532.33 eV compared with the original HA-PAO NFMs, but no shifts were observed for the COOH signals. The similar phenomenon was also observed in other amidoxime-adsorbents of ZIF-90-AO, PAF-AOs, SiO_2 -AO, and so on (Ahmad et al., 2020; Li et al., 2020d; Mei et al., 2022). These results indicated that the adsorption of uranium on HA-PAO NFMs occurred via the chelation of uranyl ions with $\text{C}(\text{NH}_2)=\text{N}-\text{OH}$ groups of PAO rather than other groups.

2.3. Anti-biofouling ability

The anti-biofouling ability of HA-PAO NFMs was determined by the “zone of inhibition” assay and the influence of bacteria on uranium adsorption capacities. Firstly, the significant inhibition zones demonstrated the excellent anti-biofouling ability of HA-PAO NFMs. As shown in Fig. 5a, there were significant inhibition zones (15.17 ± 0.24 , 15.69 ± 0.53 , 23.53 ± 0.34 , and $12.80 \pm 0.26 \text{ mm}$, respectively) were observed around HA-PAO NFMs for the four kinds of typical bacteria (*E. coli*, *S. aureus*, *B. subtilis*, and *V. alginolyticus*, respectively). By comparison, no inhibition zones were generated by the BC and BC-PAO membranes (no antibacterial cross-linkers were contained in the fabrication process). It is reported that the antibacterial ability of Co^{2+} may be due to the disruption of bacterial membranes via six synergistic effects, including in diffusion-directed lipid-oxidation, direct interaction, reaction-oxygen species generation, cation transport interruption, chelation effects and membrane depolarization, which leads to the death of bacteria (Zhuang et al., 2012). The above results showed that the Co^{2+} supplied the nanofiber membranes with an antibacterial ability that prevented the active sites of adsorbents from being blocked by marine microorganisms subsequently affecting their adsorption capacity.

Furtherly, bacteria had little effect on uranium adsorption capacity of HA-PAO NFMs. As shown in Fig. 5b, the uranium capture amounts (284 ± 4.9 , 281 ± 7.7 , 276 ± 6.8 , and $270 \pm 14.1 \text{ mg-U/g-Ads}$, respectively) in the real seawater and the seawater containing 10^2 , 10^3 and 10^4 CFU mL^{-1} bacteria were almost as good as that ($302 \pm 8.0 \text{ mg-U/g-Ads}$) in the sterile seawater. As a control, the bacteria had obvious influence on the uranium adsorption for the BC-PAO (without antibacterial compositions) samples. Even their uranium uptake capacities reduced by 33.91% and 48.07% when they were soaked in the U-spiked seawater containing 10^3 and 10^4 CFU mL^{-1} bacteria (Fig. S12). The high performances of HA-PAO NFMs in the bacteria-contained seawater were attributed to the antibacterial properties of and the strong uranium uptake ability of amidoxime groups. Combined with the apparent inhibition zones, these results suggest that HA-PAO NFMs have excellent bacterial inhibition ability in uranium extraction from seawater.

2.4. Reusability

The reusability of HA-PAO NFMs was evaluated by the adsorption-desorption experiments, and the cycles were done ten times. As shown in Fig. S13, Fig. S14 and Fig. 5c, 91% uranium can be desorbed from the adsorbents in 35 min after being immersed in the eluent. The regenerated adsorbents still have a good uranium adsorption ability in the next adsorption experiment. After 10 cycles, the uranium capture amount of

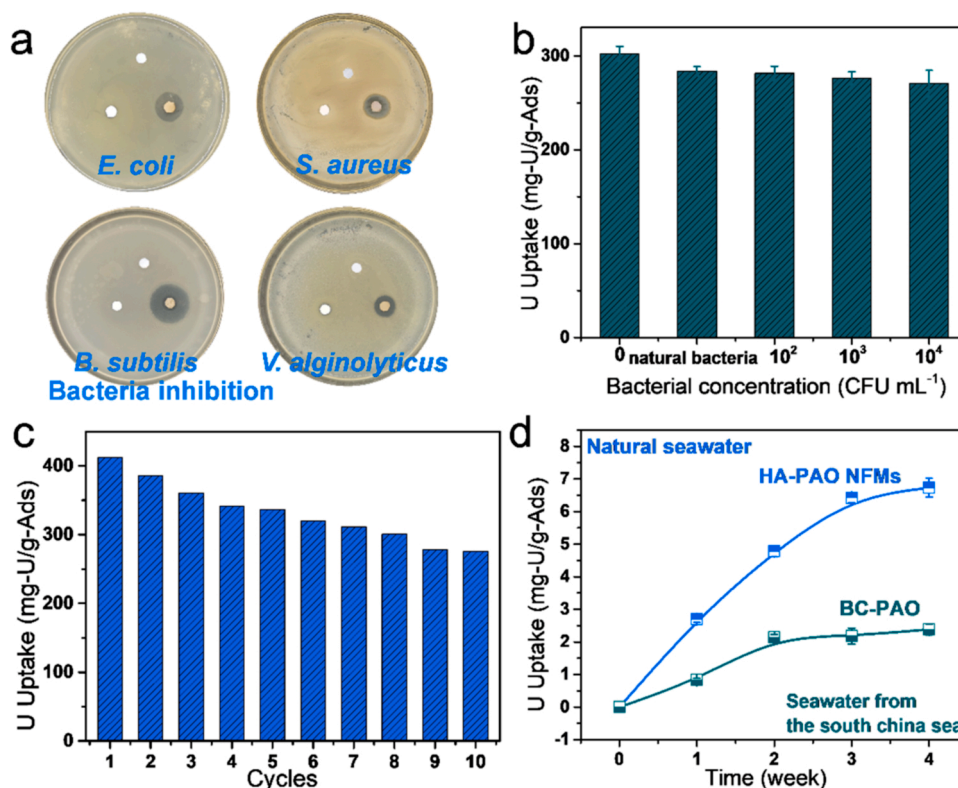


Fig. 5. a) Photos obtained from the “zone of inhibition” assays to investigate the bacteria inhibition ability of HA-PAO NFMs. As a control, the inhibition zones of the BC and BC-PAO also were tested. b) Comparison of the uranium adsorption amounts of HA-PAO NFMs in the bacteria-contained seawater and sterile seawater. c) Reusability of the HA-PAO NFMs. d) Comparison of the U-uptake capacities of HA-PAO NFMs and BC-PAO in the natural seawater.

HA-PAO NFMs still reached up to 275.55 mg-U/g-Ads in the simulated seawater, and the elution rate still can be maintained at 81.33%. The above results illustrate that the HA-PAO NFMs have a long service life in the uranium extraction, which may be benefitting from their excellent mechanical property and structural stability.

2.5. U-uptake from the natural seawater

The uranium extraction from natural seawater (100 L, obtained from South China Sea near Hainan Island) was performed to investigate the real application ability of HA-PAO NFMs. As shown in Fig. 5d, the BC-PAO (without antibacterial compositions) obtained a uranium capture amount of 2.38 ± 0.17 mg-U/g-Ads after 28 days of soaking in natural seawater. Interestingly, the HA-PAO NFMs obtained a much higher uranium capture capacity of 6.73 ± 0.29 mg-U/g-Ads than that of BC-PAO. The uranium extraction capacity of HA-PAO NFMs was 1.83 times higher than that of the adsorbents without antifouling ability. These results demonstrated the effective uranium extraction by HA-PAO NFMs and also displayed the benefits of the anti-biofouling properties for uranium extraction from natural seawater. Biofouling caused by marine microorganisms usually lowers the uranium extraction efficiency, the antifouling compositions can effectively prevent the blockage of active sites of adsorbents by the microorganisms, which can improve the uranium extraction.

2.6. Comparisons with other adsorbents

The high specific surface area of fiber adsorbents is mainly obtained through the construction of ultrathin or porous fibers. However, these methods not only lower the mechanical strength but also aggravate the complex spinning technology. Meanwhile, to withstand the complex ocean environment and realize the large-scale exploitation of uranium, the high strength and the massive fabrication of adsorbents are an

urgent need. In this work, a robust and anti-biofouling HA-PAO NFM was designed via a supramolecular crosslinking. Compared with other fibers, including those in the ORNL research team (the world-leading research group for uranium adsorption that has focused on the development of fiber adsorbents for many years), the HA-PAO NFM simultaneously achieved ultrathin diameter (20–30 nm), excellent specific surface area ($51.02 \text{ m}^2 \text{ g}^{-1}$), and high tensile strength (13.6 MPa) by the introduction of BC frameworks in the supramolecular crosslinking process. Its diameter is only 1/10–1/9000 of those of previously reported fibers, but its BET area and mechanical strength are 1.70–26.85 and 5.67–9.25 times of the reports, respectively, these properties of HA-PAO NFMs are very superior in the fiber-based adsorbents (Table S3) (Christou et al., 2019; He et al., 2020; He et al., 2021; Kaur et al., 2021; Li et al., 2020c; Oyola et al., 2016; Su et al., 2018; Wang et al., 2018).

Porous structure, effective adsorption ligands, and antifouling ability endowed the HA-PAO NFMs with very high uranium extraction capacities. Their uranium capture capacity in natural seawater was 1.83 times higher than that of non-antifouling adsorbents. In addition, they can rapidly decontaminate uranium in wastewater, and the residual uranium was reduced to as low as 7.94 ppb. Furthermore, they obtained a very high uranium capture capacity of 409 mg-U/g-Ads in the 8 ppm U-spiked simulated seawater. This adsorption amount was 1.35–20.45, 1.31–14.10, 3.20–7.72, and 1.95–2.99 times higher than those of other uranium adsorbents, including in fiber-based adsorbents, membrane-based adsorbents, gel-based adsorbents, and porous powder adsorbents, respectively (Table S4) (Bai et al., 2020; Chatterjee et al., 2015; Chen et al., 2020; Christou et al., 2019; Cui et al., 2020; Huang et al., 2020a; Kaur et al., 2021; Lin et al., 2022; Ma et al., 2019; Su et al., 2018; Sun et al., 2021; Wang et al., 2020; Xiong et al., 2019; Xu et al., 2019; Yuan et al., 2020a; Yuan et al., 2021; Zhong et al., 2020). The enhanced uranium capture ability is attributed to their porous structure, excellent hydrophilicity, massive and tight immobilization of adsorption ligands, and good antifouling ability (Fig. 6).

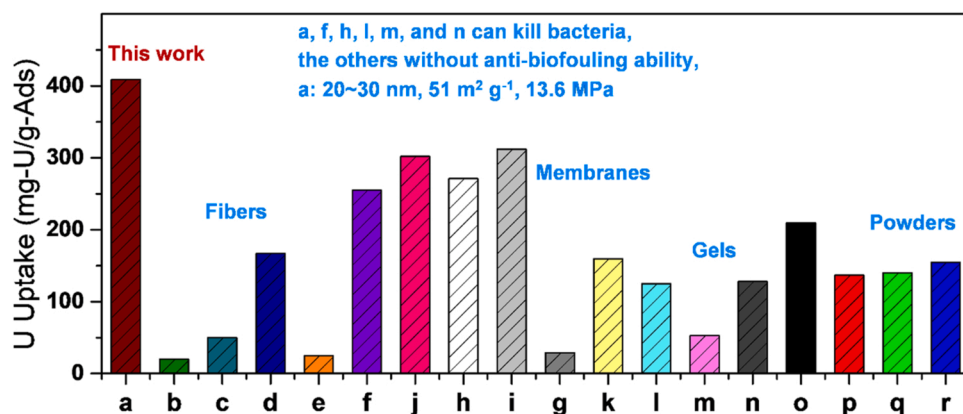


Fig. 6. Comparisons of the uranium uptake capacity, antifouling ability, fiber diameter, BET area, and mechanical strength of the HA-PAO NFMs with different types of uranium adsorbents (including fibers, membranes, gels, and porous powders). a [This work], b (Kaur et al., 2021), c (Chatterjee et al., 2015), d (Christou et al., 2019), e (Su et al., 2018), f (Huang et al., 2020a), j (Xu et al., 2019), h (Sun et al., 2021), i (Ma et al., 2019), k (Chen et al., 2020), l (Lin et al., 2022), m (Bai et al., 2020), n (Yuan et al., 2021), o (Yuan et al., 2020a), p (Xiong et al., 2019), q (Zhong et al., 2020), r (Cui et al., 2020).

3. Conclusions

In summary, we explored a novel and simple method to construct nanofiber membranes for the uranium recovery from seawater and wastewater. The obtained HA-PAO NFMs consisted of ultrathin nanofibers (20–30 nm of diameter), but displayed an ultrahigh mechanical strength (13.6 MPa, a single membrane can carry a weight of 5 Kg) owing to BC frameworks. Furthermore, the membranes displayed a large BET area ($51.02 \text{ m}^2 \text{ g}^{-1}$) and rapid sink rate of water droplets (0.23 s), which can facilitate the diffusion of seawater in the NFMs. Meanwhile, a large amount of adsorption ligands (55.65 wt%) were loaded onto the membranes, endowing the membranes with excellent uranium uptake ability of 409 mg-U/g-Ads (in 8 ppm uranium-spiked simulated seawater), 6.73 mg-U/g-Ads (in the natural seawater), 99.2% of uranium removal efficiency (in 1000 ppb uranium-contaminated wastewater). Significant inhibition zones were produced around the membrane's surface, and the uranium adsorption capacities almost unaffected by the bacteria, indicating the excellent antifouling ability of the HA-PAO NFMs. During the adsorption process, a clear color change of the adsorbents can be observed by the naked eyes, and the kinetic models and XPS data verify the chemical chelation of amidoxime ligands to uranyl ions. Moreover, the membranes possess a long service life for ten times of adsorption-desorption cycles, and still can maintain a stable structure after the cycles and after one week of immersion in the simulated waves, indicating their potential application for practical uranium extraction. This robust, highly efficient, and economical nanofiber membrane would be a promising adsorbent for the industrial uranium extraction from seawater and U-contaminated wastewater.

CRedit authorship contribution statement

S. S., R. W., and N. W. conceived the research and designed the experiments. S. S., R. W., and S. M. performed the experiments, G. X., G. Y., and C. M. contributed to part of material characterization and uranium adsorption. All authors contributed to the analyses of data and the discussions. S. S., R. W., and N. W. wrote the paper. S. S., R. W., C. M., and N. W. revised the paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Statement of environmental implication

Uranium is one of the most hazardous radionuclides, and the uranium-containing wastes (discharged from the nuclear fuel cycle or nuclear accidents) have serious chemical toxicity. This work designed high-strength and anti-biofouling nanofiber membranes (HA-PAO NFMs) for the high-performance uranium recovery. The NFMs displayed high BET area, excellent mechanical strength, and good anti-biofouling ability. More importantly, they can efficiently capture uranium from the natural seawater and U-contaminated wastewater (the residual uranium can be reduced to as low as 7.94 ppb). Meanwhile, the fabrication process is economical and eco-friendly, and the NFMs have potential in the large-scale uranium recovery.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2022.128983](https://doi.org/10.1016/j.jhazmat.2022.128983).

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